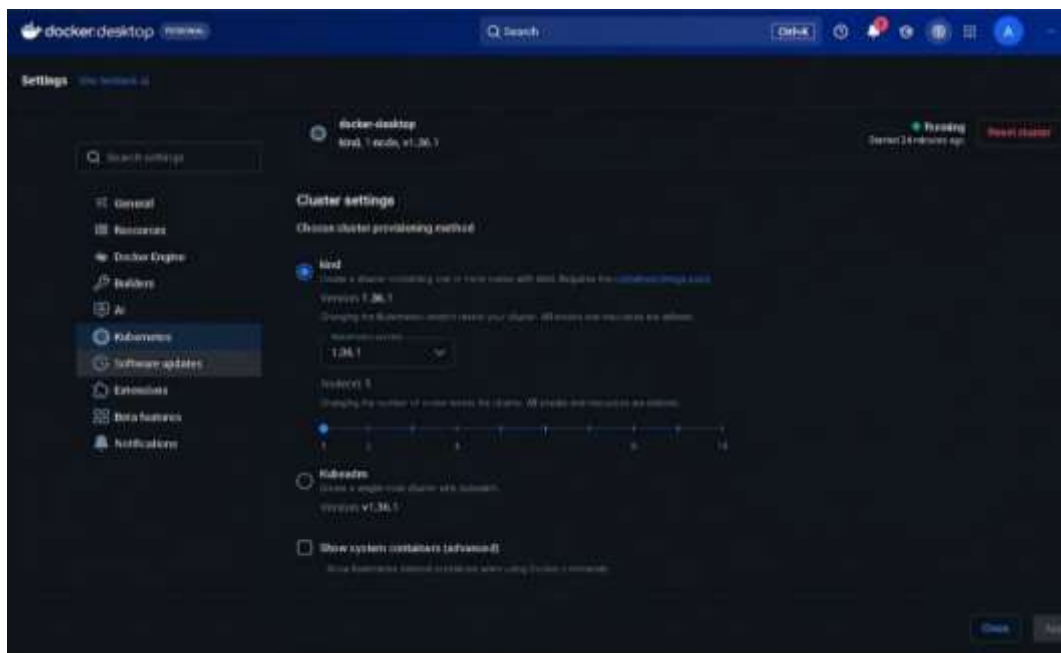
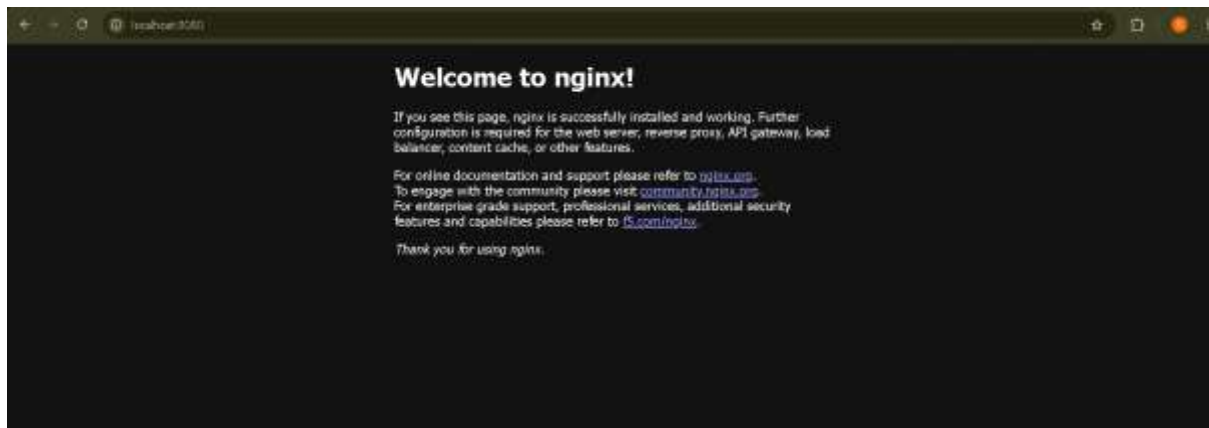


1. Deploy an application to a Kubernetes



```
ayushi2323@LAPTOP-6LVIDNI:~$  
ayushi2323@LAPTOP-6LVIDNI:~$ cd -  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl config get-contexts  
CURRENT NAME CLUSTER AUTHINFO NAMESPACE  
* docker-desktop docker-desktop docker-desktop  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl config current-context  
docker-desktop  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl get nodes  
NAME STATUS ROLES AGE VERSION  
desktop-control-plane Ready control-plane 2m12s v1.36.1  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl create deployment web --image=nginx  
deployment.apps/web created  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl get deployments  
NAME READY UP-TO-DATE AVAILABLE AGE  
web 0/1 1 0 10s  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl get pods  
NAME READY STATUS RESTARTS AGE  
web-7887448d46-7nhmM 1/1 Running 0 20s  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl scale deployment web --replicas=3  
deployment.apps/web scaled  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl get pods  
NAME READY STATUS RESTARTS AGE  
web-7887448d46-7nhmM 1/1 Running 0 41s  
web-7887448d46-fhdss 1/1 Running 0 10s  
web-7887448d46-w692x 1/1 Running 0 10s  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl expose deployment web --type=NodePort --port=80  
service/web exposed  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl get services  
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE  
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3m34s  
web NodePort 10.96.163.123 <none> 80:32207/TCP 8s  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl get endpoints web  
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 Endpoints  
NAME ENDPOINTS AGE  
web 10.244.0.5:80,10.244.0.6:80,10.244.0.7:80 15s  
ayushi2323@LAPTOP-6LVIDNI:~$ kubectl port-forward service/web 8080:80  
Forwarding from 127.0.0.1:8080 -> 80  
Forwarding from [::1]:8080 -> 80  
Handling connection for 8080  
Handling connection for 8080  
Handling connection for 8080  
Handling connection for 8080
```



```
ayushi2323@LAPTOP-6LVIDNHT:~$ kubectl delete pod <web-7887448d46-7nhm4>
-bash: syntax error near unexpected token 'newline'
ayushi2323@LAPTOP-6LVIDNHT:~$ kubectl delete pod web-7887448d46-7nhm4
pod "web-7887448d46-7nhm4" deleted from default namespace
ayushi2323@LAPTOP-6LVIDNHT:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-7887448d46-fhdss                1/1     Running   0           3m41s
web-7887448d46-tvzxf                1/1     Running   0           10s
web-7887448d46-w692x                1/1     Running   0           3m41s
ayushi2323@LAPTOP-6LVIDNHT:~$ kubectl scale deployment web --replicas=1
deployment.apps/web scaled
ayushi2323@LAPTOP-6LVIDNHT:~$ kubectl delete service web
service "web" deleted from default namespace
ayushi2323@LAPTOP-6LVIDNHT:~$ kubectl delete deployment web
deployment.apps "web" deleted from default namespace
ayushi2323@LAPTOP-6LVIDNHT:~$
```

```
ayushi2223@LAPTOP-6LVIDN  x  +  -  □
GNU nano 8.7.1 deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

ayushi2323@LAPTOP-6LVIDNHT:~$ nano deployment.yaml
ayushi2323@LAPTOP-6LVIDNHT:~$ |
```

```
ayushi2323@LAPTOP-6LVIDNHT  x  +  -  □
GNU nano 8.7.1 service.yaml *
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  type: NodePort
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30080

ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ nano service.yaml
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ |
```